More training data is still the lever

d256 x 80000 solves 0.9960 of the shared 1500-instance pool,
up from 0.9847, paired McNemar exact p = 0.0015.

Slurm 48912548 (train, 5:33) + 48999377 (eval + probes, 10:27)

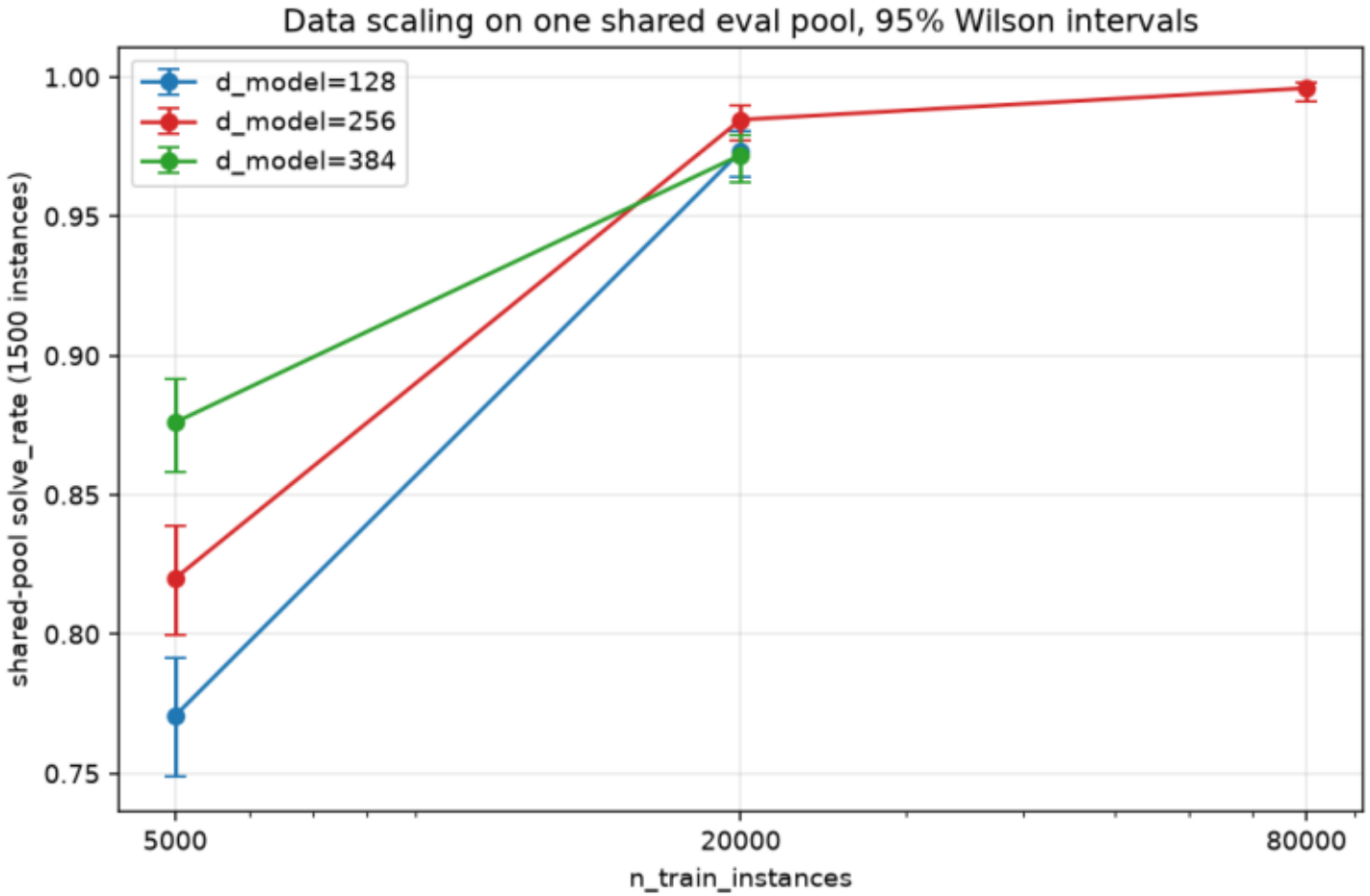
spin_gnn, branch sudoku-dd | 2026-10-22 | metric of record: shared-pool solve_rate, 1500 instances, seed_start 5e10

Seven checkpoints, one eval pool

model	shared-pool solve_rate	95% Wilson	delta vs d256 x 20000	McNemar exact p	discordant won/lost
d256 x 80000	0.9960	[0.9913, 0.9982]	+0.0113	0.00151	22/5
d256 x 20000	0.9847	[0.9771, 0.9898]	+0.0000	1	0/0
d128 x 20000	0.9733	[0.9639, 0.9804]	-0.0113	0.0241	17/34
d384 x 20000	0.9720	[0.9624, 0.9792]	-0.0127	0.0127	17/36
d384 x 5000	0.8760	[0.8584, 0.8917]	-0.1087	2.91e-38	12/175
d256 x 5000	0.8200	[0.7997, 0.8386]	-0.1647	8.38e-66	7/254
d128 x 5000	0.7707	[0.7487, 0.7912]	-0.2140	2.11e-88	6/327

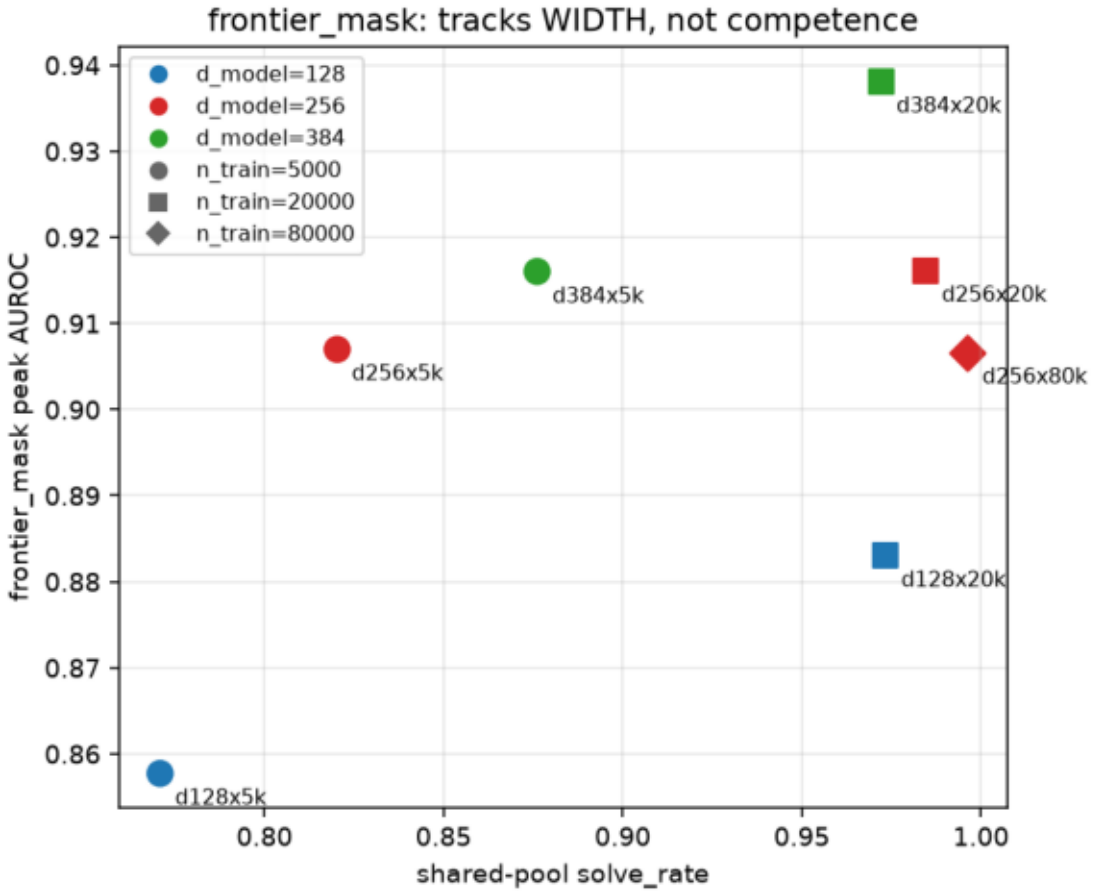
Every row is the same 1500 held-out instances, so the comparison is paired: only the instances where two models disagree carry information, and the test is an exact binomial on those. The six previously scored checkpoints reproduced their recorded rates to four decimals, which is the reproducibility check for the pool.

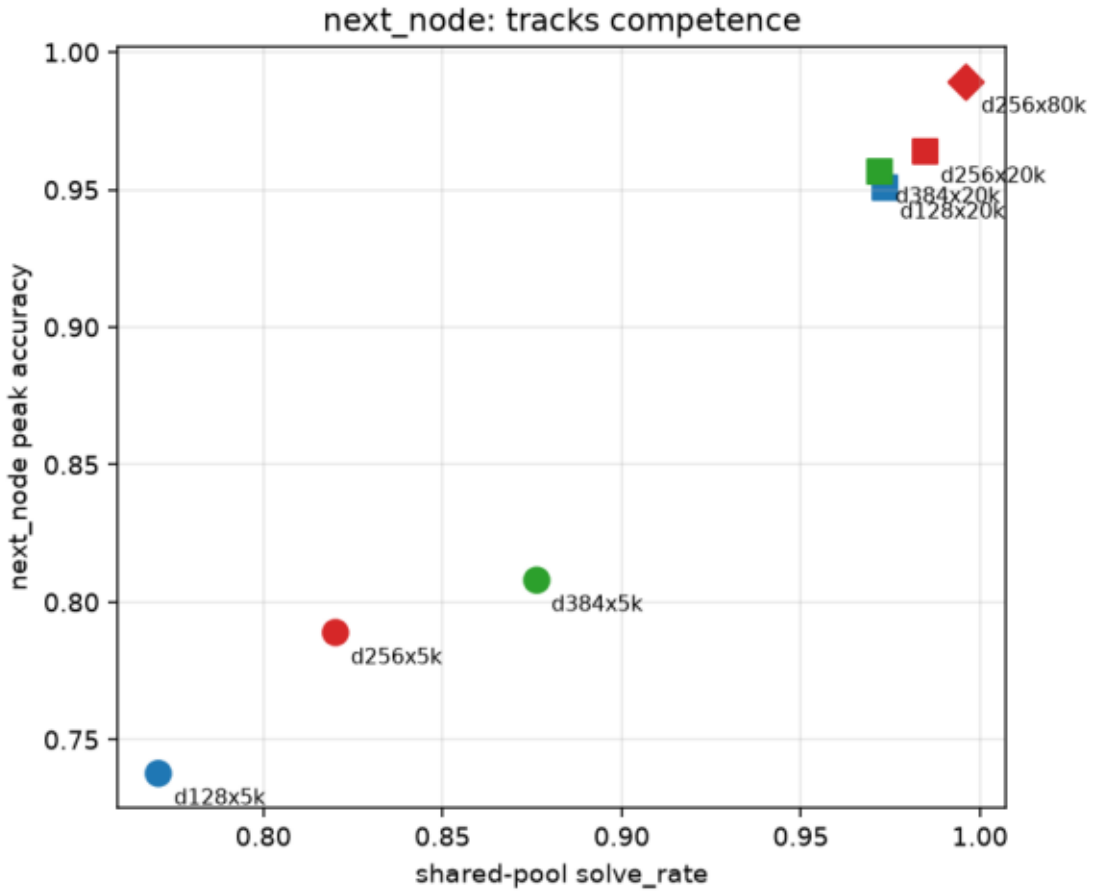
Data scaling has not saturated



Each width improves monotonically with training data and the best cell is the largest data budget tried. d0128 x 20000 (4.0M parameters) reaches 0.9733, statistically indistinguishable from d0384 x 20000 (16.7M, 0.9720) -- at this scale the data axis buys more than the width axis.

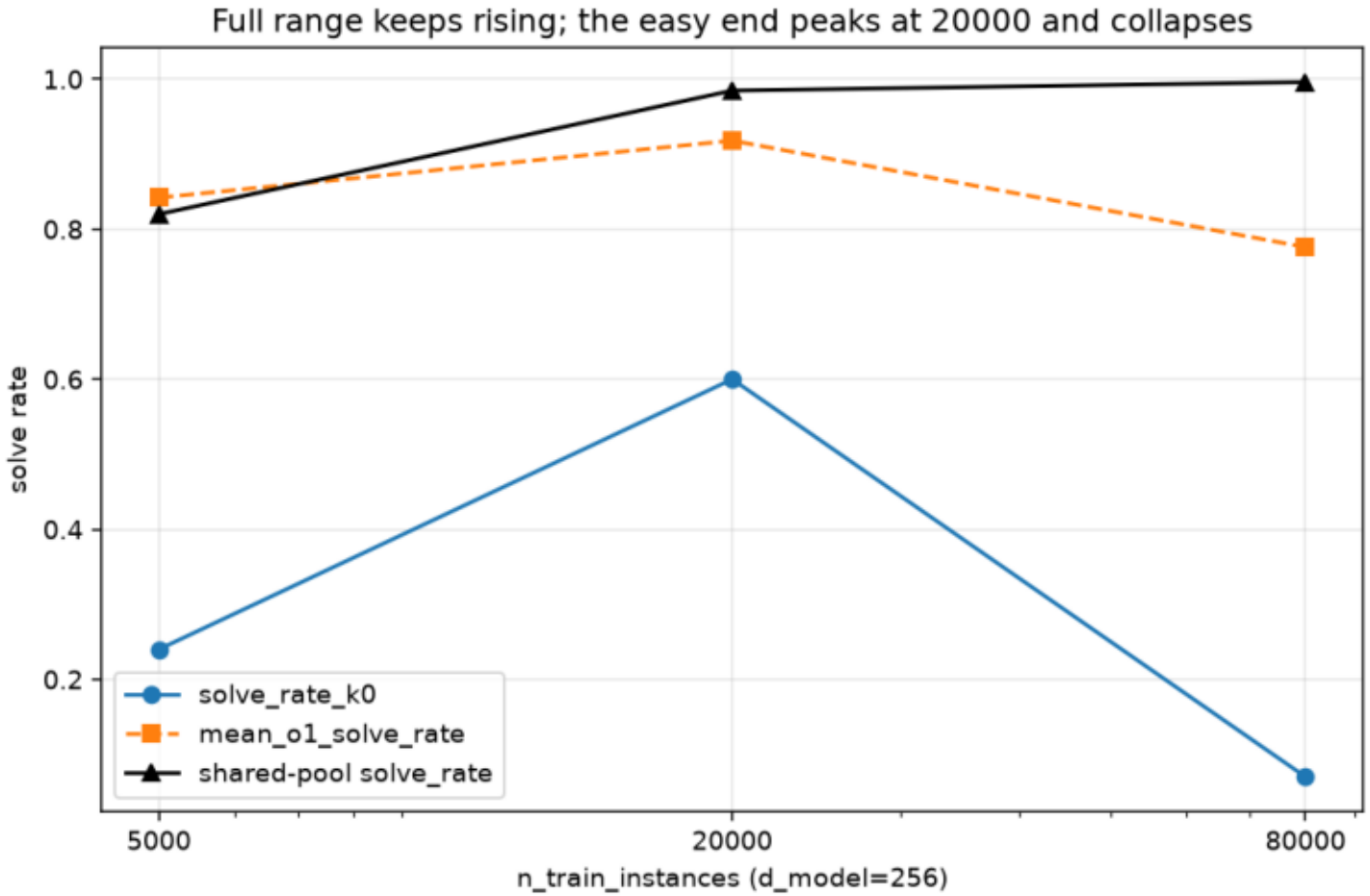
The frontier probe does not measure competence





Spearman against shared-pool solve_rate is +0.96 for next_node peak accuracy and +0.25 for frontier_mask peak AUROC; frontier_mask against d_model is +0.85. The new best model's frontier AUROC FELL (0.916 -> 0.907) while its solve rate rose. Frontier representation strength is a function of width; next-step prediction is what tracks solving.

What more data costs: the trivial cases



solve_rate_k0 is the pure-copy case (no uncolored nodes, no reasoning needed). It rose 0.24 -> 0.60 from 5000 to 20000 instances and then collapsed to 0.07 at 80000, while the full range kept rising. More width did the same thing on 10-18, so this is not specific to the data axis. These are 100-instance own-val bins and therefore noisy, but 0.60 -> 0.07 is not noise.

Pre-registration scorecard and what is next

```
HELD     solve_rate >= 0.9847 with a positive delta  ->  0.9960, +0.0113, p = 0.0015
HELD     next_node peak accuracy >= 0.96 (80000 run)  ->  0.9895
HELD     frontier AUROC for d0128 x 20000 in [0.855, 0.895]  ->  0.8831
HELD     control: d0128 x 5000 re-probe = 0.863 +/- 0.015  ->  0.8578, no drift
MISSED   predicted range [0.980, 0.995] for the 80000 run  ->  0.9960, just above
MISSED   k0 >= 0.85 for the 80000 run  ->  0.07, opposite direction
MISSED   mean_o1 >= 0.96  ->  0.776
```

Next: push the data axis again (d0256 x 320000) now that 80000 did not saturate;
find out whether the easy-end collapse is a decoding failure or a representation
failure, since k0 = 0.07 with solve_rate 0.996 is a contradiction worth resolving;
and attribute generation failures by where the first error lands (finding 5).